



DPUSCNXCameraPos

DPUSCNXCameraPos allows to control a camera view . There are two possibilities to control a view.

The first method uses hedgehogs that are triggered by a vehicle (e.g. the EGO-vehicle).

The second method allows to control a view by DPU variables.

In both cases a position relative to the vehicle is possible as well as a static position for example at a junction.

The outputs of an instance of DPUSCNXCameraPos are typically connected to an instance of DPUSGView.

1. Inputs:

<i>Family</i>	Name of the hedgehog family providing a view position (default = „ReplayPosition“)
<i>Enable</i>	=1 (default): DPUSCNXCameraPos is activated =0: DPUSCNXCamerPos is deactivated
<i>NonStaticPitchRollDeactivate</i>	For relative positioning the pitch and yaw angle of a vehicle can be ignored in order to keep the camera calm especially in a higher position. (deactivated = 1.0 (default), activated = 0.0)
<i>Hedgehog</i>	The possibilities mentioned above can be chosen by this parameter. (default = 1: positioning via hedgehogs).
<i>HHX, HHY, HHZ, HHStatic, HHLookX, HHLookY, HHLookZ</i>	See below in the description of operation mode: variables.

Inputs with signals that are normally directly connected to DPUSGView:

<i>XIn</i>	see documentation of DPUSGView
<i>YIn</i>	see documentation of DPUSGView
<i>ZIn</i>	see documentation of DPUSGView
<i>CamAln</i>	see documentation of DPUSGView
<i>CamBln</i>	see documentation of DPUSGView
<i>CamBetaln</i>	see documentation of DPUSGView
<i>CamGammaln</i>	see documentation of DPUSGView
<i>CamDeltaln</i>	see documentation of DPUSGView
<i>CamSIn</i>	see documentation of DPUSGView
<i>YawIn</i>	see documentation of DPUSGView
<i>PitchIn</i>	see documentation of DPUSGView



<i>RollIn</i>	see documentation of DPUSGEView
<i>PitchRoadIn</i>	see documentation of DPUSGEView
<i>RollRoadIn</i>	see documentation of DPUSGEView

2. Outputs:

<i>Active</i>	=0: DPUSCNXCameraPos is inactive and variables of DPUSGEView are just passed through =1: DPUSCNXCameraPos is active and controls the view via hedgehogs or DPU variables
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Output for DPUSGEView:

<i>XOut</i>	see documentation of DPUSGEView
<i>YOut</i>	see documentation of DPUSGEView
<i>ZOut</i>	see documentation of DPUSGEView
<i>CamAOut</i>	see documentation of DPUSGEView
<i>CamBOut</i>	see documentation of DPUSGEView
<i>CamBetaOut</i>	see documentation of DPUSGEView
<i>CamGammaOut</i>	see documentation of DPUSGEView
<i>CamDeltaOut</i>	see documentation of DPUSGEView
<i>CamSOut</i>	see documentation of DPUSGEView
<i>YawOut</i>	see documentation of DPUSGEView
<i>PitchOut</i>	see documentation of DPUSGEView
<i>RollOut</i>	see documentation of DPUSGEView
<i>PitchRoadOut</i>	see documentation of DPUSGEView
<i>RollRoadOut</i>	see documentation of DPUSGEView

3. Functional principle:

There are two different operation modes of this DPU. On the one hand there is the view control via hedgehog and on the other hand the view control via DPU variables.

Operation mode: Hedgehog

In this operation mode a hedgehog of the family *ReplayPosition* is used to change the position of a view. A hedgehog defines either a static position or a relative position depending on the entry *Static* in the hedgehog.

1 - Static view position (Static = 1.0)

This method is only possible for Area2 sections.

The following values of the hedgehog define the static view position:

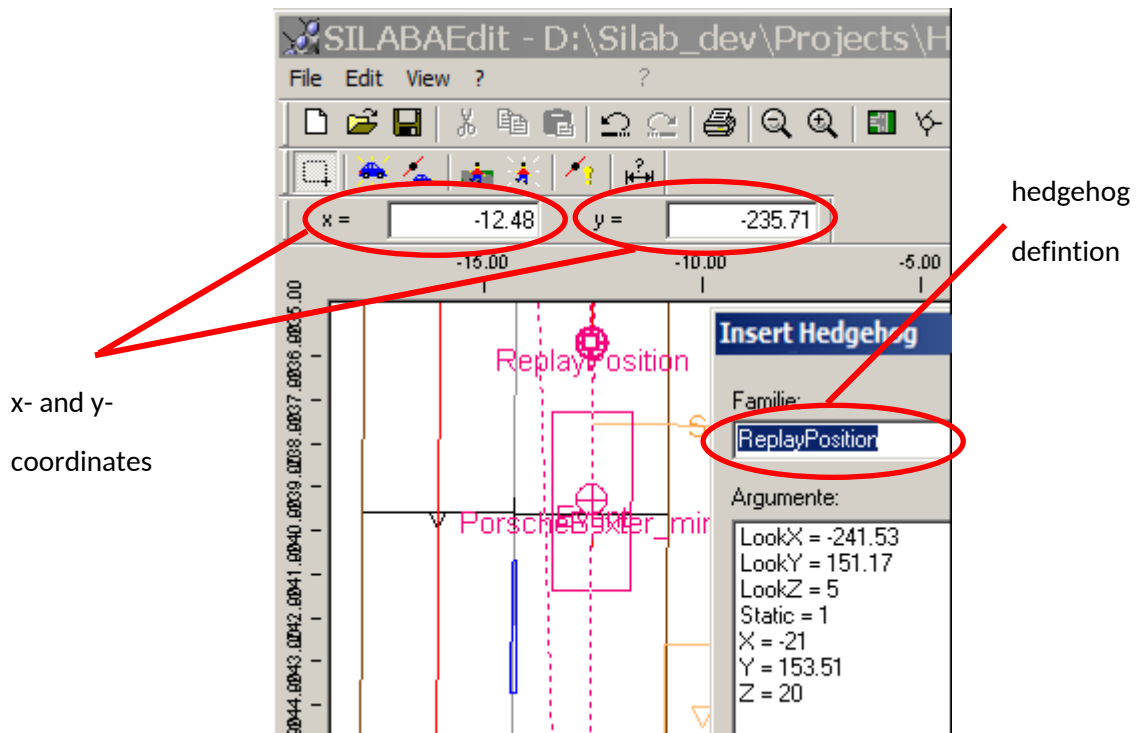
<i>X</i>	Vector (X, Y, Z) specifies the point of the camera within an area. A description for determining this vector within SILABAEEdit can be found below
<i>Y</i>	
<i>Z</i>	
<i>LookX</i>	The vector (<i>LookX</i> , <i>LookY</i> , <i>LookZ</i>) describes the direction that the camera is looking to.
<i>LookY</i>	
<i>LookZ</i>	



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Determining the coordinates in SILABAEEdit:



The z coordinate is defined by the user arbitrarily. Value *Z* as well as *LookZ* are both applied to the current height of the area.

2 - Relative view position (Static = 0.0)

This method can be applied to Area2 sections as well as to Course sections.

The following values specify a relative position:

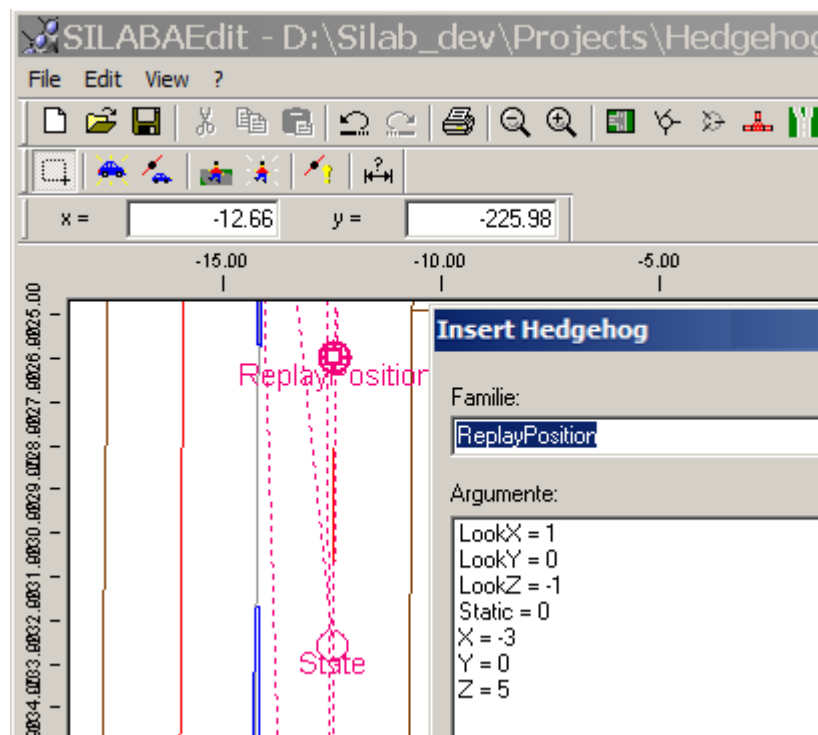
<i>X</i>	Vector (<i>X</i> , <i>Y</i> , <i>Z</i>) describes a camera position relative to the position and direction of a vehicle.
<i>Y</i>	
<i>Z</i>	
<i>LookX</i>	Vector (<i>LookX</i> , <i>LookY</i> , <i>LookZ</i>) describes the vector the camera is looking at. The direction is based on the orientation of the vehicle.
<i>LookY</i>	
<i>LookZ</i>	

Example for a relative view within an area:



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The camera position is defined 3 meters behind the vehicle and 5 meters above. The vector the camera is looking at (1, 0, -1) and defines an angle of 45 degrees downward looking at the vehicle.

If the input *Enable* is false (0.0), the DPUSCNXCameraPos stops to control the view and passes all inputs over to the outputs.

In case of a module change a static view will be disabled.

Operation mode: Variable

In that operation mode a view is controlled by a set of DPU variables. The following table shows these variables, their default value and the name of the hedgehog variable that is controlled by the variable (see operation mode hedgehog above).

Variable name	Default	Parameter in hedgehog mode
HHX	0.0	X
HHY	0.0	Y
HHZ	0.0	Z
HHStatic	0.0	Static
HHLookX	0.0	LookX
HHLookY	0.0	LookY
HHLookZ	0.0	LookZ